

# LORRE LI

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## EDUCATION

### University of Notre Dame, College of Engineering

B.S., Double Major in Computer Science, Applied Mathematics and Statistics

Notre Dame, IN

Aug 2022 - May 2026

- **GPA:** 3.97/4.00; **Honors:** Summa Cum Laude; Dean's List; The Grand Challenge Scholarship
- **Awards:** n8n AI Workflow Hackathon, 1st Place; **Activities:** Quant Trading Club Board, CS for Good, Data Science Club, SIBC
- **Teaching Assistant:** Data Structures, Theory of Computing, and Linear Algebra; supported 100+ students

## EXPERIENCE

### Superpose

San Jose, CA

Software Engineer

May - Aug 2026

- Owned technical scoping of a **multi-GPU inference scheduler** with one worker per GPU, **CUDA\_VISIBLE\_DEVICES** pinning, first-free-wins acquisition, and configurable queue/reject backpressure; replaced a serial queue through a **zero-downtime cutover**.
- Deployed one serving stack across a **six-GPU on-prem node, RunPod A100/H100, and AWS EC2**, adapting operational workflows while synchronizing **LoRA registries and GPU telemetry** across environments.
- Partnered an **SGLang inference service** for agent workloads, owning model bring-up through release: **containerized GPU deployment**, OpenAI-compatible API and structured tool-call validation, **concurrency/load testing**, and rollback readiness.
- Resolved a production blocker by eliminating a client **retry storm** and tracing failures through **FastAPI, GPU workers, and ComfyUI** using **AWS SSM and journalctl**.

### Meta

Menlo Park, CA

Software Engineer

May - Aug 2025

- Rebuilt Instagram camera **AR effect initialization infra** using **Swift** and **Objective-C++**, lifting pass rate under cold-start conditions to 99.9% and lowering warm-up hitches by 50%, using **Xcode GPU Frame Capture** to pinpoint and verify fixes
- Refactored the **Instagram camera pipeline** in the **iOS AR SDK** to a modular, interface-driven architecture, cutting **effect-apply latency by 27%** while sustaining a 60 FPS, and simplifying cross-team feature onboarding process
- Co-designed an **AR effects coordinator** with protocol-driven interfaces for gesture, audio, and **real-time image processing**; migrated the legacy OneCamera stack to **Optic** for a new AR filter factory across Meta's Family of Apps.
- Partnered with the Images team to ship **Metal/GLSL compute shaders** for separable Gaussian and color transforms

### Radical AI

New York, NY

Full-stack Software Engineer & AI Lab Assistant

Jul - Sep 2024

- Shipped an AI tutor for technical studying and onboarding training using **React.js** and **Node.js** in a team of 10
- Optimized backend **REST** paths and caching for real-time interactions, reducing querying latency and server response times by 25%
- Initiated AI models on Google **Kubernetes Engine** with **Cloud SQL**, engaging in full-stack integration using **cloud infrastructure**
- Investigated **Silent-Data-Corruption** (SDC) issue and protocolled a checksum method to detect and abort tainted runs

### CSE Department, University of Notre Dame

Notre Dame, IN

Research Assistant, LLM Systems

Sep 2024 - May 2026

- Built an **LLM quiz agent** over TedED transcripts with prompt chaining, **self-verification**, and **embedding retrieval**.
- Built a repeatable **LLM evaluation loop** with held-out prompts, failure analysis, and traceable outputs for generation ranking.
- Deployed async **Python/FastAPI APIs** with batching, caching, rate limits, and tracing, reducing latency by **25%**.

## PROJECTS

**Robot Vision Copilot** | *PyTorch, VLA, FastAPI, OpenCV/YOLO, React, ROS 2, Docker*: LoRA fine-tuned OpenVLA-7B on 500 LIBERO demos

across 10 tasks to 65% success; built an OpenCV/YOLO API with agentic ROS 2 failure replanning

**AgenticVBench** | *Python, Harbor, Docker*: Contributing evals for 100 multimodal video-agent tasks with deterministic/VLM verifiers

**Harbor** | *Python, asyncio, Daytona*: Merged ATIF trajectory metadata; authored sandbox lifecycle and reliability fixes across 5 upstream PRs

**Legal Consultant** | *Python, PostgreSQL*: Built a RAG agent with schema-constrained reasoning, retrieval, and source-grounded responses

## SKILLS

**Languages:** Python, C/C++, Swift, Java, SQL, JavaScript, Shell

**Frameworks:** PyTorch, SGLang, FastAPI, Pydantic, asyncio, ComfyUI, React, Node.js

**Tools:** Linux, Git, Docker, Kubernetes, AWS (EC2/S3/SSM), RunPod, Harbor, Daytona, CUDA, Claude Code, MCP